Homework: Routing & Application Architecture

Extend your existing React project using React Router, nested routes, dynamic URLs, and lazy loading.

You may use your existing API or the Fake API, which supports search, filtering, pagination, and fetching items by ID.

Required Routes

Route	Page
/	Home
/about	About
/items	Items List
/items/:id	Item Details
/dashboard	Dashboard
/login	Login
*	Not Found (404)

*Replace **items** with a name appropriate for your project.*

Tasks

1. Architecture & Routing

- Use react-router, RootLayout, NavBar, and Outlet.
- Implement nested routes and keep App.jsx focused on routing.
- Separate pages, components, services, hooks, and styles.
- Move API requests into services/service.js.

2. Home, About & Login

- Home: short introduction, image, and navigation links.
- About: reuse your about me page or create separate about project page.
- Login: styled placeholder for future authentication.

3. Items List

- Display items using reusable Card components.
- Implement search, **2+ filters**, sorting, and pagination.
- Synchronize filters with URL parameters using useSearchParams.
- Handle loading, errors, and empty results.

4. Item Details

- Use useParams and fetch the item by ID from the API.
- Display **7+ meaningful details**.
- Add Back and All Items links.
- Handle loading, errors, missing items, and stale requests.

5. Dashboard

- Display **4+ statistics** calculated from API data.
- Add **2+ data breakdowns** (category, status, rank, etc.).
- Implement **1+ interactive filter** affecting the statistics.
- Add links to relevant items or pages.

6. Lazy Loading & Not Found

- Lazy-load **3+ pages** using React.lazy and Suspense.
- Add a custom 404 page with a Home link.
- Handle invalid item IDs appropriately.

Requirements

- Preserve all existing project functionality.
- Navigation must work without full page reloads.
- Support direct URLs, browser Back/Forward, and page refresh.
- Use real API responses (fake api responses count if they behave like api), not hardcoded data.
- Maintain consistent, responsive styling.

Submission:

1. Attach links and press Turn In before the deadline
    a. GitHub repository link,
    b. deployed application URL

Documentation: React Router · useSearchParams · React.lazy · Suspense